Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/22/2005
PI Name:	Lupton, Joanne R. Ph.D.		
Project Title:	Nutritional Countermeasures to Radiation Exposure		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Nutrition, Physical Fitness, and Rehabilitation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-2253	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
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No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	10
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Braby, Leslie (Texas A&M University) Chapkin, Robert (Texas A&M University) Ford, John (Texas A&M University) Turner, Nancy (Texas A&M University) Carroll, Raymond (Texas A&M University)		
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Task Description:	1. Original Aims. The overall goal of this research program was to develop nutrition countermeasures to radiation-induced colon tumorigenesis, using male Sprague Dawley rats as a model system. Superimposed on the background of irradiation with Fe-ions or no irradiation was the injection of a known colon specific carcinogen, azoxymethane (AOM) in order to simulate the potential exposure to environmental contaminants. The diet interventions tested were combinations of a lipid component (fish oil vs corn oil) and a fiber component (pectin vs cellulose). At the end of the three year period we anticipated knowing: (1) if radiation exposure synergistically enhances colon tumor induction by AOM; (2) which diet combination(s) are protective against colon cancer and if this effect is due to less DNA damage, greater removal of DNA-adducted cells by apoptosis or greater repair of DNA-adducted cells; (3) if short term studies (e.g. initiation or aberrant crypt formation) are predictive of later tumor development; and (4) if noninvasive technology can be used to detect specific mRNAs that are predictive for radiation exposure and/or response to that exposure, which would have later application to humans. 2. Key Findings. The preliminary project results demonstrated that our selected radiation dose and sampling times were appropriate for the proposed experimental design. We discovered that prior exposure to radiation before exposure to a chemical carcinogen increased the severity of the preneoplastic lesions formed during colon tumorigenesis. Expression of genes associated with response to carcinogen exposure, as well as control of cell cycle kinetics is influenced by radiation exposure, above and beyond that observed with the chemical carcinogen. The dietary manipulations demonstrate that fermentable fiber and n-3 fatty acids: 1) influence the redox state of colonocytes in irradiated rats and alter the redox response to subsequent carcinogen exposure; 2) affect the level of oxidative adducts present in DNA after carcinogen exposure; 3) are able to reduce the formation of preneoplastic lesions of colon cancer through changes in cell cytokinetics regulated by altered gene expression; and 4) reduced the incidence of colon tumor incidence. 3. Impact of Findings on Project Goals. The findings from our experiment indicate diet is capable of influencing the initiation, promotion and final tumor stage in colon cancer. Therefore, diet may serve as a viable countermeasure to help protect against radiation-enhanced colon cancer. 4. Proposed Research Plan for the Coming Year. With the termination of this project we are initiating the work proposed in the new project. However, we will continue to evaluate the data collected in order to further elucidate the likely relationships between the responses at the initiation and promotion stages as predictors of the final tumor outcomes. This will include relationships between the gene expression data to be collected in the new project and the phenotypic observations collected during this project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Information generated from our research project will easily extend to Earth medical research. Individuals exposed to radiation on Earth, either through their profession, or for diagnostic or therapeutic radiation procedures will benefit from the knowledge gained. Our results, and the questions derived from them, should provide avenues for research to address the types of diets that might enhance the risk associated with professional exposure to radiation. It will also determine if there are dietary recommendations that would protect individuals from the increased risk occurring from radiation exposure when it is used for diagnostic techniques. Another area of potential application includes developing dietary recommendations that would benefit individuals undergoing radiotherapy as a cancer treatment by increasing the susceptibility of the cancer cells to the treatment, while helping to protect the normal cells.
Task Progress:	During the last year we completed the live animal component of the project, finalized tumor characterization, and completed the remaining laboratory assays. There were several key findings. At the initiation stage of colon tumorigenesis fish oil supplementation results in less oxidative DNA damage to colon cells than does corn oil as evidence by lower levels of 8-OHdeoxyguanosine levels. We had previously shown lower levels of methylation damage to colon DNA. One mechanism by which fish oil elicits this response is by producing a more reduced colonic environment, so that when there is an oxidative insult, the apoptosis of DNA damaged cells is initiated. These changes in redox state over time were tracked by measuring the ratio of reduced to oxidized glutathione. At the promotion stage of colon tumorigenesis (aberrant crypt formation) the combination of fish oil and pectin results in a lower number of aberrant crypts and a lower number of high multiplicity aberrant crypts (those considered most predictive of future tumor development. Apoptosis also plays an important role at this stage of tumorigenesis with pectin resulting in a greater proportion of cells undergoing apoptosis than cellulose (P=0.02). Radiation-induced changes in gene expression profiles (measured by microarray from mRNA of exfoliated cells)were observed for proteins involved in signal transduction (insulin receptor, PPARgamma coactivator 1,and prostaglandin E receptor 1) and antioxidant function (coenzyme Q). At the final tumor stage (31 weeks after the 2nd carcinogen injection) rats fed corn oil were 2.2 times more likely to have a tumor than those fed fish oil (P=0.02). Collectively, these data show that a specific diet intervention can be an effective countermeasure against radiation-enhanced colon tumorigenesis.
Bibliography Type:	Description: (Last Updated: 08/22/2020)
Articles in Peer-reviewed Journals	Apanasovich TV, Sheather S, Lupton JR, Popovic N, Turner ND, Chapkin RS, Braby LA, Carroll RJ. "Testing for spatial correlation in nonstationary binary data, with application to aberrant crypt foci in colon carcinogenesis." Biometrics. 2003 Dec;59(4):752-61. PubMed PMID: 14969452 , Dec-2003
Articles in Peer-reviewed Journals	Bancroft LK, Lupton JR, Davidson LA, Taddeo SS, Murphy ME, Carroll RJ, Chapkin RS. "Dietary fish oil reduces oxidative DNA damage in rat colonocytes." Free Radic Biol Med. 2003 Jul 15;35(2):149-59. PubMed PMID: 12853071 , Jul-2003
Articles in Peer-reviewed Journals	Hong MY, Chapkin RS, Barhoumi R, Burghardt RC, Turner ND, Henderson CE, Sanders LM, Fan YY, Davidson LA, Murphy ME, Spinka CM, Carroll RJ, Lupton JR. "Fish oil increases mitochondrial phospholipid unsaturation, upregulating reactive oxygen species and apoptosis in rat colonocytes." Carcinogenesis. 2002 Nov;23(11):1919-25. PMID: 12419841 , Jan-2003
Articles in Peer-reviewed Journals	Hong MY, Chapkin RS, Davidson LA, Turner ND, Morris JS, Carroll RJ, Lupton JR. "Fish oil enhances targeted apoptosis during colon tumor intiation in part by down regulating Bc1-2." Nutr Cancer. 2003;46(1):44-51. PubMed PMID: 12925303 , Oct-2003

Articles in Peer-reviewed Journals	Sanders LM, Henderson CE, Hong MY, Barhoumi R, Burghardt RC, Wang N, Spinka CM, Carroll RJ, Turner ND, Chapkin RS, Lupton JR. "An increase in reactive oxygen species by dietary fish oil coupled with the attenuation of antioxidant defenses by dietary pectin enhances rat colonocyte apoptosis." J Nutr. 2004 Dec;134(12):3233-8. PubMed PMID: 15570018 , Dec-2004
Articles in Peer-reviewed Journals	Sanders LM, Henderson CE, Hong MY, Barhoumi R, Burghardt RC, Carroll RJ, Turner ND, Chapkin RS, Lupton JR. "Pro-oxidant environment of the colon compared to the small intestine may contribute to greater cancer susceptibility." Cancer Lett. 2004 May 28;208(2):155-61. PubMed PMID: 15142673 , May-2004
Articles in Peer-reviewed Journals	Turner ND, Braby LA, Ford J, Lupton JR. "Opportunities for nutritional amelioration of radiation-induced cellular damage." Nutrition. 2002 Oct;18(10):904-12. Review. PMID: 12361786 , Jan-2003
Articles in Peer-reviewed Journals	Wu G, Fang YZ, Yang S, Lupton JR, Turner ND. "Gluthathione metabolism and its implications for health." J Nutr. 2004 Mar;134(3):489-92. Review. PubMed PMID: 14988435 , Mar-2004
Awards	Lupton J. "Dr. Lupton received the Dannon Institute Mentorship Award, January 2004." Jan-2004
Awards	Lupton J. "Dr. Lupton was appointed Lifetime Associate, the National Academy of Sciences, January 2004." Jan-2004
Presentation	69) Lupton, J. R., N. D. Turner, N. Popovic, R. S. Chapkin, L. A. Braby, J. R. Ford, and R. J. Carroll "Nutrition as a possible countermeasure to radiation exposure " Jan-2003
Presentation	78) Chapkin, R. S., J. R. Lupton, N. D. Turner and L. A. Davidson "Changes in gene expression in exfoliated cells: role of bioactive food components " May-2003
Presentation	Bancroft, L. K., J. R. Lupton, S. S. Taddeo, L. A. Davidson, M. E. Murphy, R. J. Carroll, and R. S. Chapkin "Fish oil protects against intestinal oxidative DNA damage " Apr-2002
Presentation	Henderson, C. E., L. M. Sanders, M. Y. Hong, S. S. Taddeo, C. M. Spinka, N. D. Turner, R. J. Carroll, R. S. Chapkin, and J. R. Lupton "Colonocyte DNA damage differs with diet but not age " Apr-2002
Presentation	Lupton, J. R., N. D. Turner, N. Popovic, R. S. Chapkin, L. A. Braby, and J. R. Ford "Nutritional countermeasures to radiation exposure " Aug-2002
Presentation	79) Hong, M. Y., L. K. Bancroft, N. D. Turner, R. S. Chapkin, M. E. Murphy, R. J. Carroll, and J. R. Lupton "Differential responses to oxidative DNA damage may explain the difference in cancer susceptibility between small (SI) and large intestine (LI) " Jun-2003
Presentation	80) Sanders, L. M., C. E. Henderson, M. Y. Hong, R. Barhoumi, R. C. Burghardt, M. E. Murphy, R. J. Carroll, N. D. Turner, R. S. Chapkin, and J. R. Lupton "A diet of fish oil and pectin reduces colonic antioxidant enzymes creating an environment permissive for ROS-induced apoptosis " Jun-2003
Presentation	81) Newton, A. H., K. L. Covert, R. Barhoumi, N. D. Turner, L. M. Sanders, M. Y. Hong, S. S. Taddeo, C. M. Van Velson, M. E. Murphy, R. S. Chapkin, and J. R. Lupton "Dietary fish oil + butyrate protect against colon cancer by increasing reactive oxygen species and decreasing mitochondrial membrane potential in the promotion stage of tumorigenesis " Jun-2003
Presentation	Bancroft, L. K., R. S. Chapkin, L. A. Davidson, M. E. Murphy, R. J. Carroll, and J. R. Lupton "Response to oxidative stress may not explain the differential cancer susceptibility of small and large intestine " Apr-2002
Presentation	Covert, K. L., A. H. Newton, N. D. Turner, L. M. Sanders, N. Popovic, C. M. Van Velson, S. S. Taddeo, M. Y. Hong, M. E. Murphy, R. S. Chapkin, and J. R. Lupton "The combination of dietary fish oil and butyrate decreases high-multiplicity aberrant crypt foci in experimentally induced colon cancer " Apr-2003
Presentation	Hong, M. Y., L. K. Bancroft, N. D. Turner, L. A. Davidson, R. S. Chapkin, M. E. Murphy, R. J. Carroll, and J. R. Lupton "Dietary fish oil enhances targeted apoptosis response to dextran sodium sulfate (DSS)-induced oxidative DNA damage in the top part of the colonic crypt " Apr-2003
Presentation	Hong, M. Y., L. K. Bancroft, N. D. Turner, R. S. Chapkin, C. M. Spinka, R. J. Carroll, and J. R. Lupton "Dietary fish oil, compared to corn oil, decreased oxidative DNA damage at the base and middle region of the colonic crypt in dextran sodium sulfate (DSS)-treated rats " Apr-2004
Presentation	Hong, M. Y., L. K. Bancroft, N. D. Turner, R. S. Chapkin, M. E. Murphy, R. J. Carroll, and J. R. Lupton "Differential response to dextran sodium sulfate (DSS)-induced oxidative DNA damage in rat small (SI) and large intestine (LI) " Apr-2003
Presentation	Hong, M. Y., N. D. Turner, M. E. Murphy, R. J. Carroll, R. S. Chapkin and J. R. Lupton "Relationship among proliferation, p27, differentiation and apoptosis measured by colocalization in the same cell as a function of carcinogen injection " Apr-2004
Presentation	Hong, M. Y., N. D. Turner, S. S. Taddeo, M. E. Murphy, R. J. Carroll, R. S. Chapkin, and J. R. Lupton "Influence of dietary lipid and butyrate on the coordination of proliferation differentiation and apoptosis in the same cell during the initiation stage of colon carcinogenesis " Apr-2004
Presentation	Mann, J. C., N. Popovic, S. S. Taddeo, M. M. Murphy, R. S. Chapkin, N. D. Turner, and J. R. Lupton "The effects of diet and ionizing radiation on AOM-induced colon carcinogenesis " Apr-2004
Presentation	Newton, A. H., K. L. Covert, R. Barhoumi, N. D. Turner, L. M. Sanders, M. Y. Hong, S. S. Taddeo, C. M. Van Velson, M. E. Murphy, R. S. Chapkin, and J. R. Lupton "Dietary fish oil + butyrate protect against colon cancer by increasing reactive oxygen species and decreasing mitochondrial membrane potential in the promotion stage of tumorigenesis " Apr-2003

Presentation	Newton, A. H., N. D. Turner, M. Y. Hong, L. A. Davidson, L. M. Sanders, R. S. Chapkin, M. E. Murphy, R. J. Carroll, and J. R. Lupton "Dietary fish oil and butyrate may protect against colon cancer by inducing mitochondria-dependent apoptosis in the promotion stage of carcinogenesis " Apr-2004
Presentation	Popovic, N., N. D. Turner, L. A. Braby, R. S. Chapkin, and J. R. Lupton "Combined effect of radiation and a chemical carcinogen on the formation of aberrant crypt foci in rat colon " Apr-2002
Presentation	Popovic, N., N. D. Turner, L. A. Davidson, L. A. Braby, J. R. Ford, R. J. Carroll, R. S. Chapkin, and J. R. Lupton "Dietary prevention of colon cancer induced by combined effect of carcinogen and radiation " Apr-2003
Presentation	Sanders, L. M., C. E. Henderson, M. Y. Hong, R. Barhoumi, R. C. Burghardt, C. M. Spinka, N. Wang, R. J. Carroll, N. D. Turner, R. S. Chapkin, and J. R. Lupton "Dietary fish oil and pectin protect against oxidative DNA damage in rat intestinal epithelial cells due to heightened apoptosis induced by reactive oxygen species " Apr-2004
Presentation	Sanders, L. M., C. E. Henderson, M. Y. Hong, R. Barhoumi, R. C. Burghardt, M. E. Murphy, R. J. Carroll, N. D. Turner, R. S. Chapkin, and J. R. Lupton "A diet of fish oil and pectin reduces colonic antioxidant enzymes creating an environment permissive for ROS-induced apoptosis " Apr-2004
Presentation	Sanders, L. M., C. E. Henderson, M. Y. Hong, R. Barhoumi, R. C. Burghardt, R. J. Carroll, N. D. Turner, R. S. Chapkin, and J. R. Lupton "Dietary alteration of reaction oxygen species (ROS) and antioxidant enzyme activity in rat colonocytes is age-dependent " Apr-2004
Presentation	Turner, N. D., N. Popovic, J. C. Mann, W. Fu, L. A. Davidson, R. B. Isett, R. J. Carroll, L. A. Braby, J. R. Ford, R. S. Chapkin, and J. R. Lupton "Simulated galactic cosmic radiation enhances colon cancer in the rat, which is mediated by changes in gene expression that are influenced by dietary intervention " Oct-2004
Presentation	Turner, N. D., N. Popovic, M. Y. Hong, S. S. Taddeo, L. A. Davidson, L. A. Braby, J. R. Ford, Q. Zheng, D. V. Nguyen, R. J. Carroll, R. S. Chapkin, and J. R. Lupton "Radiation enhances azoxymethane-induced colon cancer development " May-2004
Presentation	Turner, N. D., N. Popovic, S. S. Taddeo, L. M. Sanders, K. J. Paulhill, J. C. Mann, L. A. Braby, J. R. Ford, R. S. Chapkin, L. A. Davidson, Q. Zheng, R. J. Carroll, and J. R. Lupton "High energy LET radiation as a risk modifier in colon carcinogenesis " Oct-2003
Presentation	Zhou, G.-D., J. R. Lupton, N. D. Turner, R. S. Chapkin, and K. C. Donnelly "Endogenous DNA adducts (I-compounds) in colon and duodenum of rats treated with azoxymethane " Apr-2002